TECHNICAL ASSESSMENT REPORT: FLOOD INUNDATION MAPPING

SUBMISSION TO: GalaxEye Recruitment Task for the post of Imagery Analyst

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1. EXECUTIVE SUMMARY

This report details the development of automated change detection workflow carried out for rapid response flood detection analysis using multisensory optical and radar data. The objective was to delineate flood inundation extents by overcoming the limitations of optical sensors during cloud-heavy weather events. By fusing Sentinel-1 Synthetic Aperture Radar (SAR) with Sentinel-2 Optical imagery, the workflow successfully isolated seaonal floodwaters from permanent water bodies, providing a reliable rapid assessment product.

2. STUDY AREA

The study was conducted over an isolated area in Morigaon district of Assam, which experienced flood inundations during the second wave of monsoon in the month of July 2024.

3. OBJECTIVES

The task achieved the following objectives:

i). Script based remote sensing data download with the help of python scripts and open APIs. The Copernicus Data Space Ecosystem API was used to download the sentinel 2 image and ASF data search API was used to facilitate Sentinel-1 SAR image.

ii). Pre-processing the data to coregsiter the multi-sensor optical and SAR data, clipping the data with respect to the AOI and the generating indices such as NDWI and SAR db as input to change analysis.

iii). Conducting change detection analysis to identify and visualise the flood inundation over the region of interest.

4. DATASET SPECIFICATIONS

The analysis leveraged the complementary strengths of active and passive remote sensing:

i) Pre-Event Data (March,2024): Sentinel-2 (Multispectral Optical) : Used to establish a high-confidence baseline of permanent water bodies. The Green (Band 3) and NIR (Band 8) channels were utilized to calculate the Normalized Difference Water Index (NDWI), ensuring accurate separation of existing river channels from the flood mask.

ii) Post-Event Data (July,2024): Sentinel-1 (C-Band SAR) : Sentinel 1 SAR was selected for its all-weather capability. During flood events, cloud cover typically renders optical data unusable. Sentinel-1's C-band microwave signal penetrates clouds and rain, allowing for reliable surface monitoring.

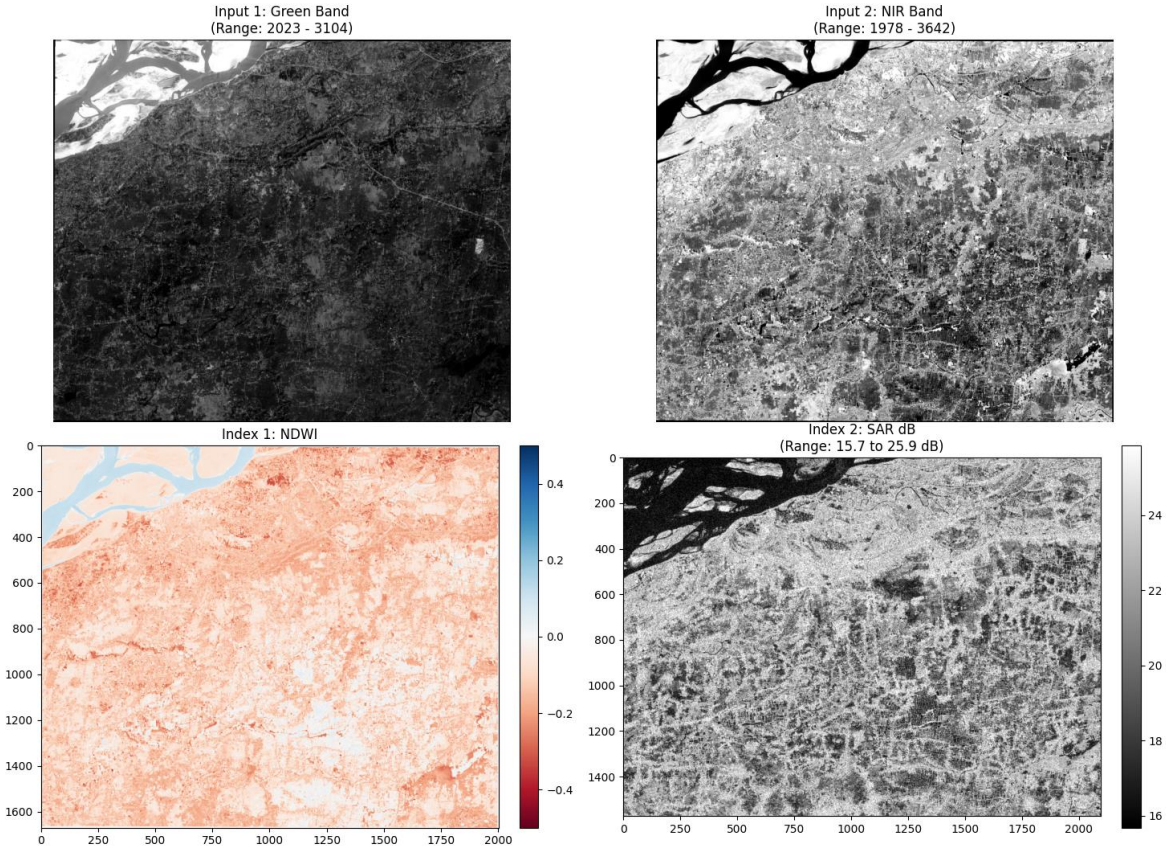

Fig1. Input data visualisation

5. METHODOLOGY

A hybrid "Change Detection by Masking" approach was implemented to minimize false alarms:

1. **Speckle Filtering:** A 5x5 Median Blur was applied to the SAR intensity data. This non-linear filter was chosen to reduce the inherent "salt-and-pepper" granular noise of coherent radar systems while preserving the sharp edges of flood boundaries.
2. **Adaptive Thresholding:**

   i)SAR: Otsu's Binarization was applied to the inverted SAR image. This dynamic method calculates the optimal threshold by minimizing intra-class variance, making the system robust to varying incidence angles and surface wetness without manual tuning.

   ii)Optical: A fixed NDWI threshold (>0) extracted permanent water bodies.

3. **Sensor Fusion:** The final flood mask was derived by subtracting the *Optical Permanent Water Mask* from the *SAR Total Water Mask*. This ensures that the reported area represents only *new* inundation.
4. **Morphological Cleaning:** A connected-component filter removed noise clusters smaller than 2 hectares, ensuring the final output focused on significant flood zones.

5. Results

The results of the analysis identified combined fluvial and pluvial flooding in the region of interest using the automated change detection algorithm approach. The flood inundation statistics shows that 6784 hectares of land to be inundated in the region. This shows the intensity of flood severity over the state of Assam and the capability of automated remote sensing workflow in rapid response Disaster management and mitigations.

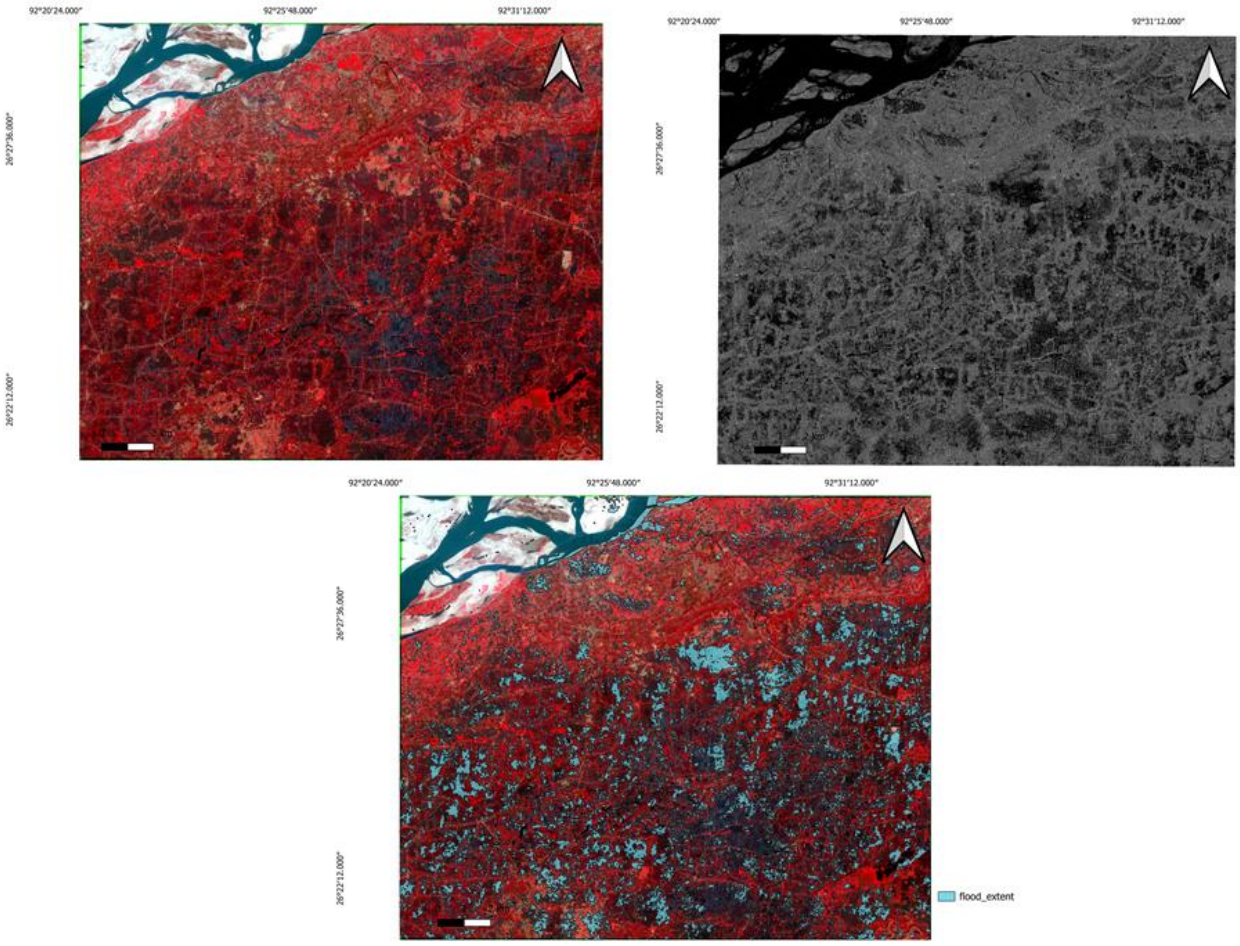

Fig 2. Flood inundation detection

Limitations

The workflow has several areas of improvements such as use of advanced indices and machine learning based classification over simple indices to further enhancing the change analysis by refining the filtering approach to reduce false alarms which has not been addressed due to time constraints.